

Question ID: 9f70fd47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

What is the y -coordinate of the y -intercept of the graph of $\frac{3x}{7} = -\frac{5y}{9} + 21$ in the xy -plane?

Correct Answer: 189/5, 37.8

Rationale

The correct answer is $\frac{189}{5}$. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, which is a point with an x -coordinate of 0 . Substituting 0 for x in the given equation yields $\frac{3(0)}{7} = -\frac{5y}{9} + 21$, or $0 = -\frac{5y}{9} + 21$. Subtracting 21 from both sides of this equation yields $-21 = -\frac{5y}{9}$. Multiplying both sides of this equation by -9 yields $189 = 5y$. Dividing both sides of this equation by 5 yields $\frac{189}{5} = y$. Therefore, the y -coordinate of the y -intercept of the graph of the given equation in the xy -plane is $\frac{189}{5}$. Note that $189/5$ and 37.8 are examples of ways to enter a correct answer.